BUYZA HOTEL MANAGEMENT SYSTEM (BuHoMaS).

Project Name: BUYZA HOTEL MANAGEMENT SYSTEM (CodeName: BuHoMaS).
Company, Owner, and Developer: BUYZA Inc.
Name of the Software: HotelloSys.

Project Overview

- **Software Name:** HotelloSys
- **Type:** Desktop enterprise Hotel Management System (HMS)
- **Purpose:** Centralized system to manage all hotel operations.
- **Modules:**
    1. **Room Reservations** – Booking and managing rooms.
    2. **Restaurant** – Food & drink orders.
    3. **Bar** – Drinks only.
- **Functionalities Required:**
    - Inventory management (food, drinks, etc.)
    - Employee management (concierges, waiters, chefs, etc.)
    - Billing and invoicing across modules
    - Customer management with ranking and point system
    - Reports: employee-level (daily, weekly, monthly), hotel-level (weekly, monthly, quarterly, semi-annual, annual)
    - Subscription billing management
    - Authentication: Every employee has a login and a profile
    - Admin functionality: system setup, user management, subscription tracking, maintenance

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Preliminary Thoughts

1. **Architecture**
   - This will need a **modular desktop application** architecture.
   - Backend database (likely SQLite for single hotel or PostgreSQL/MySQL for multi-hotel scalability).
   - Layered design:
     - **UI Layer** – desktop interface
     - **Business Logic Layer** – rules for billing, reports, points
     - **Data Layer** – database operations
2. **Technology Stack**
   - **Language:** I recommend **C# with .NET 7/8** (or Java with JavaFX) for desktop apps.
     - **Why C#:** Mature for desktop apps, great GUI support with WPF/WinForms, integrates easily with databases, and strong support for reporting and printing.
   - **Database:** SQLite (start), upgradeable to PostgreSQL, SQL Server, or MySQL for enterprise scaling.
   - **IDE:** Visual Studio Community Edition (for C#) or IntelliJ/NetBeans for Java.
3. **Reports and Printing**
   - We can use **Crystal Reports** or **FastReport** for invoice/billing and summary reports.
   - Daily, weekly, and monthly reports can be auto-generated from database queries.
4. **Authentication & Roles**
   - Implement **role-based authentication**: Admin vs Employee (with department-specific permissions).
   - Each employee profile tracks activity for accountability.
5. **Additional Considerations**
   - **Customer Loyalty System:** Track spending, rank customers, issue points.
   - **Subscription Module:** Track recurring payments for hotel subscriptions.

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Proposed Development Plan

Step 0: Set up Development Environment

1. Install IDE: **Visual Studio Community Edition**.
2. Install .NET SDK (latest stable version).
3. Install database: **SQLite** + GUI tool (DB Browser for SQLite).
4. Install reporting tool: **FastReport or Crystal Reports**.
5. Setup Git repository for version control.

Step 1: Project Skeleton

- Create a C# WPF Desktop Project: HotelloSys.
- Define folder structure:

  /HotelloSys

  /DataLayer

  /BusinessLogic

  /UI

  /Reports

  /Images

  /Utils
- Setup database connection.

Step 2: Authentication Module

- Employee login & registration.
- Role-based access (Admin vs Staff).
- Employee profile management (with photos).

Step 3: Core Modules Development

1. **Room Reservations**
   - CRUD operations for rooms and bookings
   - Invoice generation

2. **Restaurant**
    - Menu management
    - Orders, inventory, and billing
3. **Bar**
    - Drinks-only menu
    - Orders, inventory, and billing

Step 4: Reports

- Daily, weekly, and monthly employee reports.
- Hotel-level reports.
- Subscription invoices.

Step 5: Customer Loyalty Module

- Customer registration & profile
- Spending tracking & point system
- Ranking the best customers

Step 6: Admin Module

- System setup on install
- Manage employees & roles
- Subscription management
- System maintenance

Step 7: Testing & Deployment

- Unit testing per module.
- Integration testing (all modules interact correctly).
- Packaging for hotel installation.
- Installer creation.